Holistically Evaluating Reasoning Over Legal Texts in Large Language Models

Sophia Moore, In Submission to the Yale Department of Computer Science & Department of Mathematics
Advised by Prof. Arman Cohan, Department of Computer Science, Yale University

Context

The increasing capacities and prominence of large language models (LLMs) lends itself to applications across various industries. In particular, adopting LLMs to the legal field could increase accessibility to legal resources and alleviate lawyer burden.

The existing benchmark for evaluating an LLM's legal reasoning capacity, *LegalBench*, independently tests LLM capabilities across tasks across the four IRAC categories: issue-spotting (I), rule-recall (R), rule-application (A) & rule-conclusion (C). This misrepresents a model's true capabilities, as to produce a correct answer in a real-world scenario, the model must correctly work across all five subdivisions simultaneously. This project will explore the ways in which we can more accurately evaluate LLM capabilities in the domain of reasoning over legal texts through a more comprehensive testing approach.

Key Objectives

1. Establish a subset of open-ended, natural language tasks testing model ability to modularly and sequentially apply established legal framework reasoning (IRAC)
2. Evaluate 19 frequently-used LLMs (15 open-source & 4 commercial) to draw conclusions surrounding performance trends, parameter count variation across families, and compare of closed-API vs. open source models as it pertains to reasoning in the legal sphere
3. Highlight future directions for performance improvement via prompt-engineering and intentional legal application.

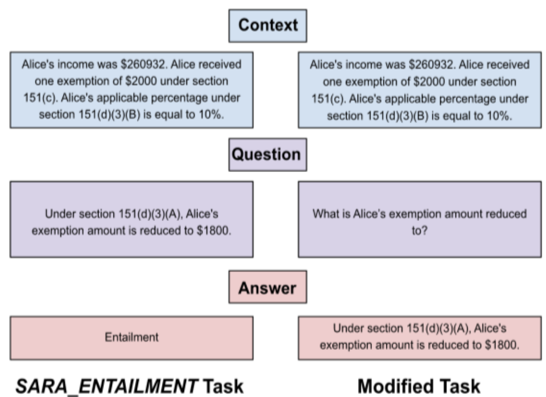
The SARA Dataset

The StAtutory Reasoning Assessment (SARA) dataset was created to investigate the effects of natural language understanding approaches on statutory reasoning. In particular, the SARA dataset consists of a set of rules extracted from the statutes of the US Internal Revenue Code (IRC), and a set of natural language questions that can be answers correctly only through interpretation of the aforementioned rules.

Importantly, key features of American law retained across SARA cases include reasoning with time, exceptions and substitutes, numerical reasoning, cross-references, common sense knowledge, and hierarchical structures. The SARA dataset itself consists of both binary prediction tasks (*sara_entailment*) and integer prediction (*sara_numeric*).

Task Construction

We collapse redundancies in the the *sara_entailment* test-split to create a subset that can be dated and reframed into open-ended questions testing sequential IRAC application. Following, *LegalBench* precedent, these tasks are intended to be evaluated as zero-shot prompts to test to test baseline rule-recall capabilities. Here, we give a basic example of how a *sara_entailment* task can be reframed:



Models are scored on the quality and correctness of their analysis:

Incorrect Analysis	Partially Correct Analysis	Correct Analysis
The relevant section of the Internal Revenue Code is Section 2(a), which defines a surviving spouse as someone who remains unmarried at the end of the taxable year following the death of their spouse. In this scenario, Alice died on July 9th, 2021, which means Bob was a surviving spouse in 2020, since he was still married to Alice at that time.	Section 2(a) of the Internal Revenue Code defines a surviving spouse as someone whose spouse has died during the tax year and has not remarried before the end of the tax year. In this scenario, Alice died on July 9th, 2014, and Bob did not remarry by the end of 2016. Therefore, Bob was a surviving spouse in 2016.	In Section 2(a), the Internal Revenue Code defines a surviving spouse as a taxpayer whose spouse died during either of the two taxable years immediately preceding the taxable year. To qualify as a surviving spouse, the person must not have remarried. In this scenario, Alice died on July 9th, 2014, and Bob did not remarry by the end of 2016. Therefore, Bob was a surviving spouse in 2016.

Evaluation Metrics

Classification tasks are evaluated using an "exact-match." To account for a label imbalance in the task subset we use balanced-accuracy as a metric.

Additionally, for a standardized comparison metric, we compute Mean Absolute Percent Error (MAPE) on numerical predictions output by models on rule-conclusion tasks.

$$\text{Balanced Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\frac{TP_i}{TP_i + FN_i} \qquad \text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100, \quad \text{for } y_i \neq 0$$

3B Parameters

	Rule-Recall	Rule-Application	Rule-Conclusion	Balanced Accuracy	MAPE
Bloom-3B	7.97	3.26	13.77	16.53	684.014%
Flan-T5-XL	5.80	4.35	47.26	65.69	226.39%
Incite-Instruct-3B	10.87	4.35	39.13	43.76	261.11%
Opt-2.7B	2.90	2.17	34.42	49.18	518.86%

7B Parameters

	Rule-Recall	Rule-Application	Rule-Conclusion	Balanced Accuracy	MAPE
Bloom-7B	13.68	6.16	23.91	31.33	178.68%
Falcon-7B-Instruct	4.71	2.90	35.50	42.74	227.81%
Incite-7B-Base	15.58	7.25	14.49	12.12	48.34%
Incite-7B-Instruct	30.07	7.25	14.49	20.14	92.02%
MPT-7B-8k-Instruct	5.07	7.61	46.38	64.17	58.17%
OPT-6.7B	5.80	3.62	13.77	14.39	247.39%
Vicuna-7B-16k	19.20	14.86	39.13	52.22	149.01%

13B Parameters

	Rule-Recall	Rule-Application	Rule-Conclusion	Balanced Accuracy	MAPE
Flan-T5-XXL	23.55	13.77	50.72	68.72	198.21%
OPT-13B	9.42	4.34	23.19	28.43	199.19%
Vicuna-13B-16k	23.19	18.84	31.88	44.74	31.19%
WizardLM-13B	7.61	10.14	38.41	41.49	33.74%

Commercial

	Rule-Recall	Rule-Application	Rule-Conclusion	Balanced Accuracy	MAPE
Claude-3.7	93.48	71.01	62.32	78.97	24.34%
GPT-4	73.19	62.32	63.04	76.87	9.23%
GPT-4o	74.64	66.00	65.22	81.42	29.85%
GPT-3.5-Turbo	50.72	42.75	52.90	62.79	18.22%

Results & Conclusions

Our results show that, generally speaking, higher parameter size correlates to higher scores across all components of the IRAC framework. Family-specific trends appear to hold across different size points, suggesting that pretraining and instruction-tuning play an important role in driving result variation amongst different model families.

We notice a substantial gap between the performances of commercial and open-source models, particularly in the areas of rule-recall and rule-application. Rule-application is the lowest-scoring category across all models, regardless of parameter count. Importantly, this indicates that chain-of-thought prompting could play a significant role in improving models' analytic capabilities.

All tested LLMs seem to perform better on the given binary tasks as opposed to the open-ended questions. Common model errors include reasoning with time, using out-of-context definitions, arithmetic mistakes, and incorrect analysis.